

SharePoint Archive Automation

PnP.PowerShell + app-only certificate auth
Full detail: Scripts-Reference.md •
SPO-Archive-Setup-Guide.md • PROJECT-HANDOFF.md

TECHNICIAN QUICK REFERENCE

1 ONE-TIME SETUP PER CLIENT TENANT (~15 minutes)

1. Install modules — PowerShell 7: `Install-Module PnP.PowerShell -Scope CurrentUser -Force`. Windows PowerShell 5.1 (separate window): `Install-Module Microsoft.Online.SharePoint.PowerShell -Scope CurrentUser -Force`
2. Register the app + certificate (PowerShell 7, browser sign-in as Global Admin):

```
Register-PnPEntraIDApp -ApplicationName "SPO-Archive-Automation" -Tenant "CLIENT.onmicrosoft.com" -SharePointApplicationPermissions "Sites.FullControl.All"
```
3. Entra portal → App registrations → API permissions → **Grant admin consent**. A 401 right after this is usually just propagation delay (5–30 min), not a real problem.
4. Confirm app-only auth isn't blocked tenant-wide (Windows PowerShell 5.1): `Connect-SPOService -Url https://CLIENT-admin.sharepoint.com` then `Get-SPOTenant | Select DisableCustomAppAuthentication` — if **True**, run `Set-SPOTenant -DisableCustomAppAuthentication $false`
5. Copy the printed **ClientId** and **Thumbprint** into a config JSON (start from `config.template.json`) with the site URLs. Leave `DryRun: true` for the first pass on any site you haven't run before.

2 THE SCRIPTS (run from the project folder, in PowerShell 7)

SCRIPT	WHAT IT DOES	EXAMPLE
Archive-SharePointFiles.ps1	Copies files past the age threshold into a separate Archive site; deletes originals only after the copy verifies. Behavior is set by the config's <code>DryRun</code> / <code>TestFileLimit</code> fields.	<code>.\Archive-SharePointFiles.ps1 -ConfigPath .\config-media.json</code>
Verify-Archive-Reconciliation.ps1	Read-only ground truth. Compares source vs. archive directly, ignoring the log entirely. Use this after any interrupted run.	<code>.\Verify-ArchiveReconciliation.ps1 -ConfigPath .\config.json</code>
Rename-ProblemFiles.ps1	Finds/renames names containing # % .. trailing periods or leading/trailing spaces — these break the copy job. Dry run by default; add <code>-Execute</code> to apply.	<code>.\Rename-ProblemFiles.ps1 -ConfigPath .\config-digital.json -Execute</code>
Archive-ToM365-ColdStorage.ps1	Alternative approach: archives files in place to Microsoft's native cold-storage tier, no site move. Needs Graph <code>Sites.ReadWrite.All</code> + <code>AllowFileArchive</code> enabled first.	<code>.\Archive-ToM365ColdStorage.ps1 -ConfigPath .\config-m365.json -WhatIf</code>

3 GOLDEN RULES

- **PowerShell edition split.** PnP.PowerShell needs PowerShell 7; the SPO module needs Windows PowerShell 5.1. They can't share one session — always use two windows.
- **Always dry-run first.** `DryRun: true` / `-WhatIf` / `-no -Execute` on any site or config you haven't run before.
- **Silence isn't broken.** Multiple full runs on one tenant can trigger throttling and go quiet for 10–15 minutes with zero output — that's retry backoff, not a hang.
- **Fix names before archiving.** Run `Rename-ProblemFiles.ps1` on a site before a full run if you haven't already — #, %, . . , and stray spaces will otherwise fail mid-run.
- **Don't trust the log alone after a crash.** Both archive scripts write their CSV log only once, at the very end — a closed window loses the audit trail even though real copies/deletes already happened. Run `Verify-ArchiveReconciliation.ps1` for ground truth instead.

4 MICROSOFT 365 ARCHIVE (“COLD STORAGE”) — GETTING A FILE BACK

Unlike the site-copy approach above, files stay at their **original URL and permissions** — only the storage tier changes, and it's cheaper. Good for cases where the same users should still be able to find and restore their own old files without anyone doing manual work.

End user	Find the file where it's always lived (it never moves) → right-click (or the ... menu) → Reactivate .
Timing	Instant if reactivated within 7 days of archiving. Up to 24h after that. Up to 48h for a large folder reactivated recursively. An email arrives when it's ready.
Admin / bulk	Microsoft Graph: <code>POST /sites/{siteId}/drive/items/{itemId}/unarchive</code>
Cooldown	Once reactivated, a file can't be re-archived again for roughly 120 days (Microsoft-imposed).

Generated from the project's working documentation — treat `SPO-Archive-Setup-Guide.md` as the source of truth if this card and the full docs ever disagree.